

D: Since Y and Z is the same distance from charge Q, the potential at both point is the same.

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Ans: B

A: $R = V/I = 12/3.0 = 4.0 \, \Omega$

B: $E = VIt = 12 \times 3.0 \times 5.0 = 180 \, J$

C: $Q = It = 3.0 \times 5.0 = 15 \, C$

D: Since potential is also defined as work done per unit charge, $[V] = V = J \, C^{-1}$.

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Ans: A